

Mohamed Emam

Biomechanics Researcher · AHPRA-Registered Physiotherapist

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PROFESSIONAL SUMMARY

Biomechanics researcher and AHPRA-registered physiotherapist completing a joint PhD at Semmelweis University (Budapest) and Cairo University (2022–2026). I bridge clinical rehabilitation with cutting-edge biomechanics using markerless motion capture (OpenCap) and statistical parametric mapping. Nine years of clinical and academic experience, 8 peer-reviewed publications in Q1/Q2 journals, and full general registration with AHPRA. Research spans gait analysis, neurorehabilitation, electrical stimulation, and aging physiology. Seeking postdoctoral fellowship opportunities.

EDUCATION

Joint PhD in Health Sciences	2022 – 2026
<i>Semmelweis University, Budapest & Cairo University, Egypt</i>	
Thesis: effect of eccentric versus concentric overload resistance training on the cost of walking, mechanical efficiency & muscle architecture in healthy older adults. Supervisors: Prof. Tibor Hortobágyi & Assoc. Prof. Magda Ramadan.	
MSc in Physical Therapy	2015 – 2019
<i>Cairo University, Egypt</i>	
BSc in Physical Therapy (Honours)	2008 – 2013
<i>Cairo University, Egypt</i>	

PROFESSIONAL EXPERIENCE

Assistant Lecturer	2016 – Present
<i>Kafr El-Sheikh University, Egypt</i>	
Teaching musculoskeletal physiotherapy, lab demonstrations, curriculum development, student research mentorship.	
Founder & Physiotherapist	2015 – Present
<i>Start-Rehabilitation Centre, Kafr El-Sheikh</i>	
Clinical management of musculoskeletal & orthopaedic conditions; manual therapy and exercise rehabilitation; clinic operations.	
R&D Team Lead	2022 – 2025
<i>Physioway (Physiotherapy Technology Company)</i>	
Directed development of clinical applications (scoliosis assessment tool, vestibular rehabilitation app).	

SELECTED PUBLICATIONS

- Exploring the Psychological Distress, Socioeconomic Hardship, and Academic Performance Among University Students During a Period of National Crisis. *Social Sciences & Humanities Open*, 2026 (accepted).
- Using Machine Learning-Based Classification of Postural Stability in Cervicogenic Headache Patients. *Life*, 16(7):1061, 2026. Q1
- The Role of Hydrotherapy in Enhancing Recovery after Knee Arthroplasty: A Systematic Review and Meta-Analysis. *Healthcare*, 14(13):2005, 2026. Q1
- Factors Influencing Health-Related Quality of Life in Children with Leukaemia in Saudi Arabia: A Cross-Sectional Study. *BMC Pediatrics*, 2026.

5. The Effect of Surface EMG Biofeedback on Dysphagia in Stroke Patients: A Systematic Review. *Developments in Health Sciences*, 2026 (accepted).
6. Impact of Russian Current Therapy Targeting the Anterior Tibial Group on Balance and Fall Risk in Post-Stroke Patients: An RCT. *Neurological Sciences*, 2026.
7. Wrist Muscle Performance in Students with Chronic Nonspecific Neck Pain: An Isokinetic Assessment. *BMC Musculoskeletal Disorders*, 2026.
8. Effect of Gaze Direction Recognition Task on Pain, ROM and Functional Activities in Cervicogenic Headache Patients. *BMC Neurology*, 2026.
9. Proprioceptive Training Reduces Headache Burden and Centre of Pressure Path Length in Cervicogenic Headache: An RCT. *Physiology International*, 2025.
0. Effect of Different Types of Electrical Stimulation on Postural Control and Gait After Stroke: A Systematic Review and Meta-Analysis. *Physiology International*, 112(4), 2025. Q2
1. Blood Flow Restriction Exercise Protocols for Elderly Populations: A Scoping Review. *Journal of Clinical Medicine*, 14(12):4185, 2025. Q1
2. Proprioceptive Training Improves Postural Stability and Reduces Pain in Cervicogenic Headache Patients: An RCT. *Journal of Clinical Medicine*, 13(22):6777, 2024. Q1
3. Effect of Cervicogenic Headache on Myoelectrical Activities of Suboccipital Muscles, ROM and Functional Activities at Different Ages. *Medical Journal of Cairo University*, 2019.

RESEARCH INTERESTS

- **Gait & movement biomechanics:** markerless motion capture (OpenCap), musculoskeletal modelling, statistical parametric mapping
- **Postural control & proprioceptive rehabilitation in cervicogenic headache:** surface EMG, posturography and machine-learning analysis of biomechanical data
- **Neurorehabilitation:** electrical stimulation, gait retraining and fall-risk reduction after stroke
- **Aging physiology:** eccentric vs. concentric resistance training, walking economy, muscle architecture

TECHNICAL SKILLS

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| ▸ Motion capture: OpenCap (markerless), marker-based systems | ▸ Imaging: MRI muscle segmentation & analysis |
| ▸ Statistics: SPM1D, Linear Mixed-Effects Models | ▸ Assessment: Isokinetic dynamometry, surface EMG |
| ▸ Programming: MATLAB, Python, R, SPSS | ▸ Design: RCTs, systematic reviews, meta-analyses |

AWARDS, REGISTRATION & LANGUAGES

- **AHPRA** — Physiotherapist, Full General Registration (No. PHY0002905130)
- **Scholarships & grants:** Stipendium Hungaricum · Global Korea Scholarship · SE 250+ Excellence PhD (2025, 2026) · EKÖP Doctoral Excellence Grant (2026) · Pannonia Scholarship (2026) · Researchers Supporting Project Grant, PNU (2025) · ECSS and Motor Control conference travel grants (2025, 2026)
- **Recognition:** Best Project Award, Biomedical Data Summer School, Semmelweis University (2026); Best Poster Award, 1st International Health Science Conference, Szeged (2025); 10 peer reviews for 6 international journals
- **Conferences:** ECSS Lausanne 2026 (poster) & Rimini 2025 (poster) · International Motor Control Conference, Amsterdam 2025 (oral) · 3rd Biomechanics Symposium, Budapest 2025 (poster) · 1st International Health Science Conference, Szeged 2025 (poster) · Biomedical Data Summer School, Budapest 2026 · ORPHEUS, Budapest 2026
- **Languages:** Arabic (native) · English (C1–C2, PTE 81) · Hungarian (A1)